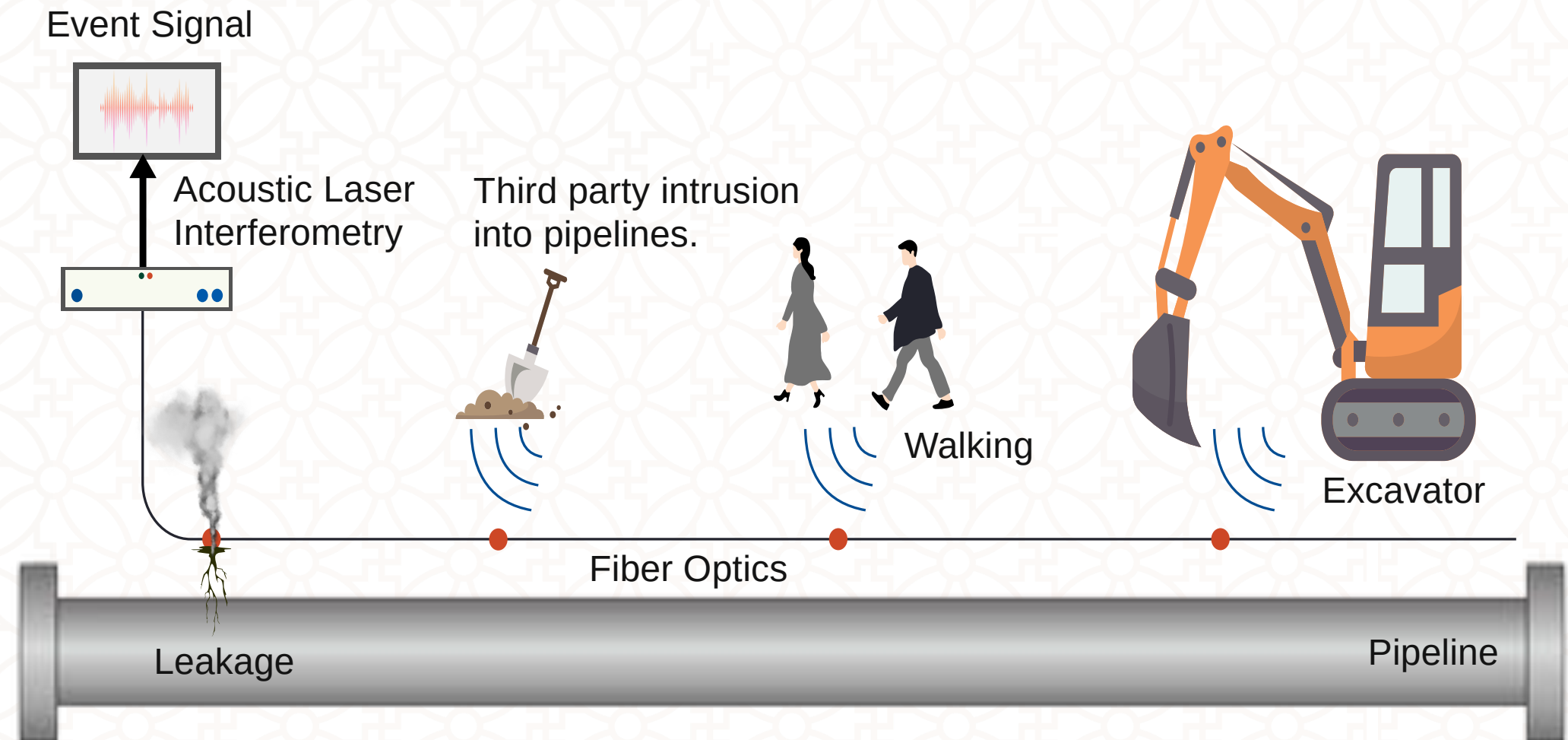


Development of Oil and Gas Pipeline Monitoring Using Fiber-Optic Distributed Acoustic Sensing and Deep-Learning-Based Signal Classification

Bagus Endar B. Nurhandoko, Yoga Hariman, Wildan Fauzi Akbar, Indriani Sukmana, Insan Rizal Komara, Wisnu Adi Guntara, R.P. Triwindhu Waskito, Elfa Fatiah, Diah Asmawati, Gunawan Handayani, Neny Kurniasih, Harry Mahardhika, Alvina K. Kuncoro, Sri Widowati, Firda Rulifiangga, Kiswanto, Fadhil Mudassir, Maria Cindrawati

Pipeline Integrity Threats

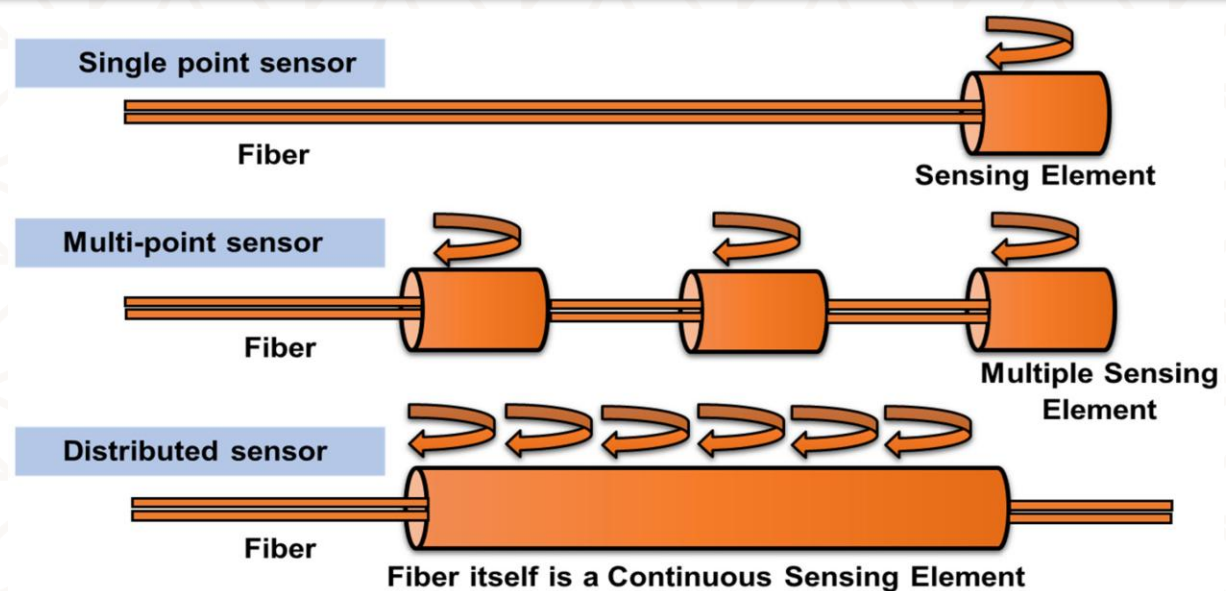
Oil and gas pipelines traverse **hundreds of kilometers**, making them highly vulnerable to **illegal tapping, third-party excavation, ground movement, and environmental fluid leaks**.



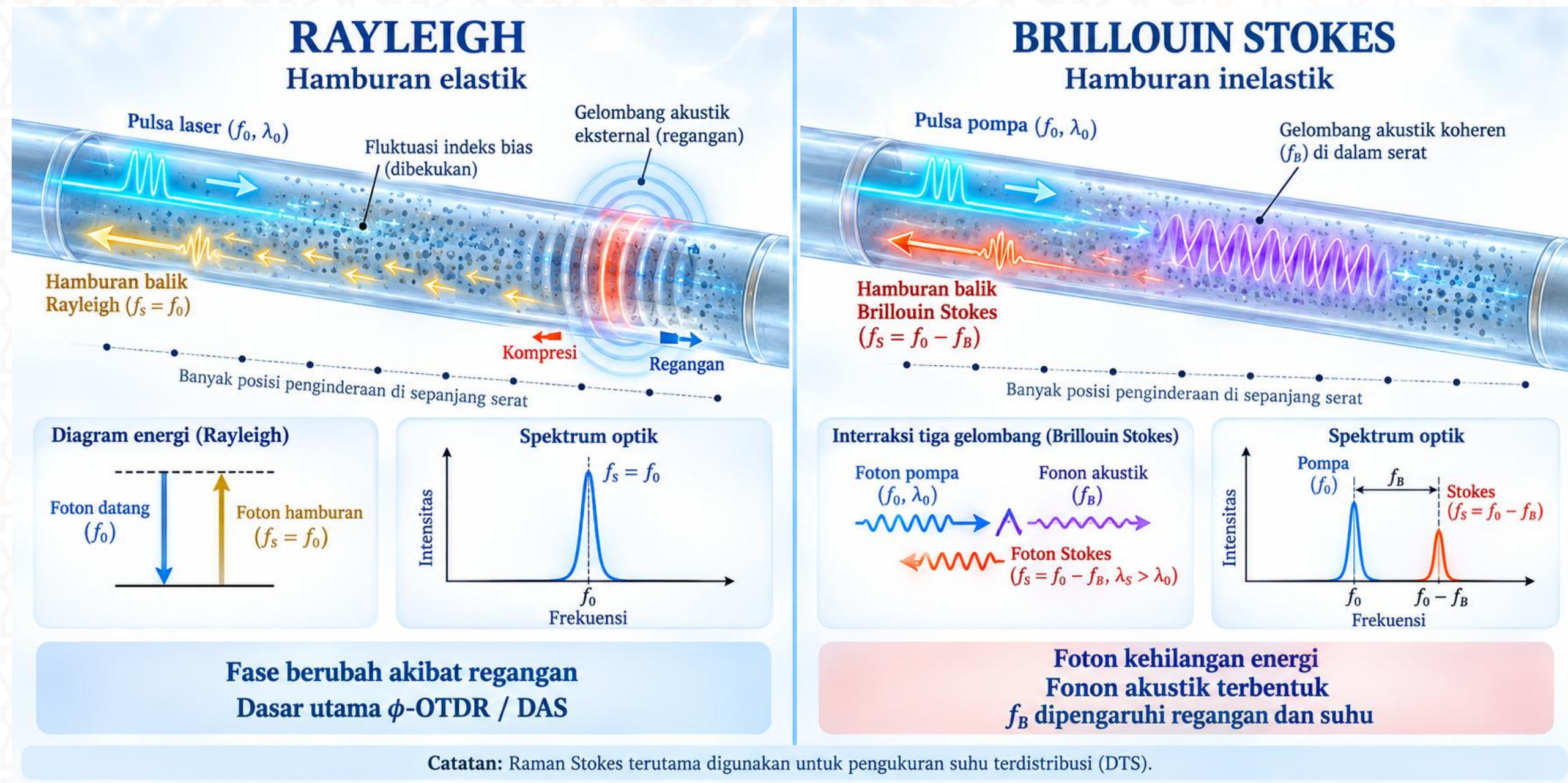
Conventional Sensor Limits



1. Pipelines often cross geographically complex and hard-to-access terrain
2. Conventional monitoring (e.g., point accelerometers, manual patrols) is difficult to implement
3. These traditional methods offer severely limited spatial coverage
4. Achieving full coverage incurs extremely high installation and maintenance costs.

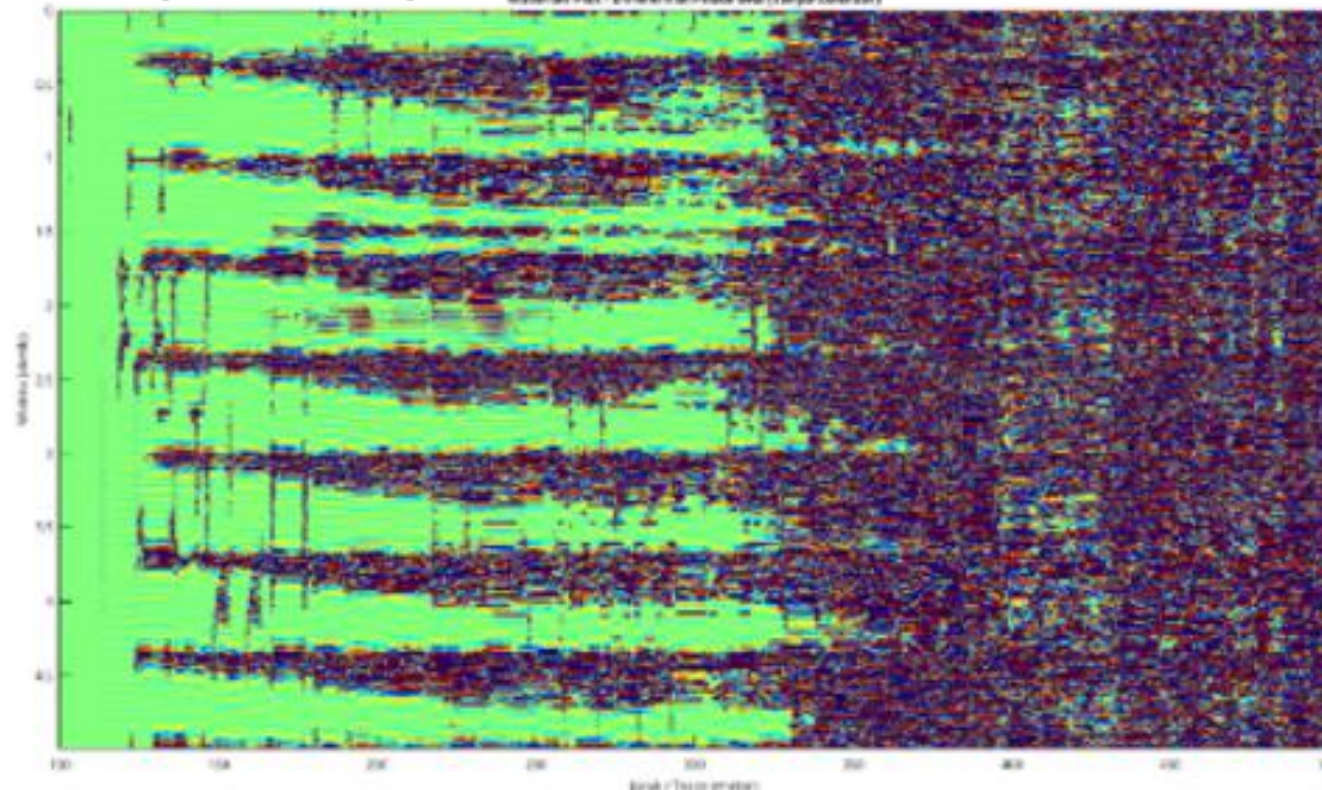


Foton Scattering in Fiber Optics Sensing

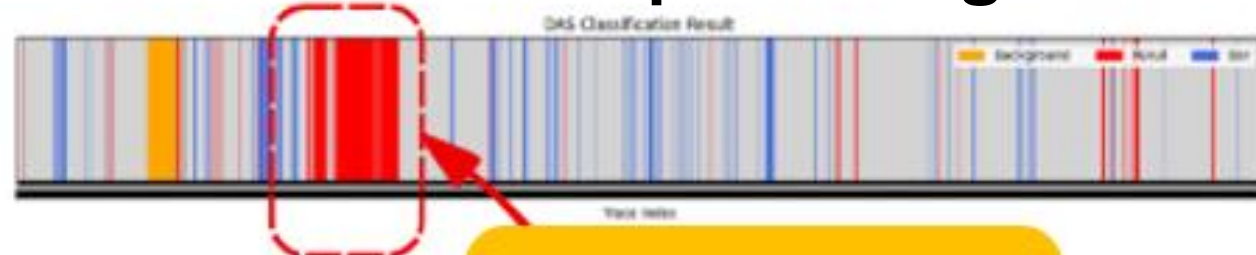


Recognition of Hammer Impact in Pipeline Using Deep Learning AI

Original Signal

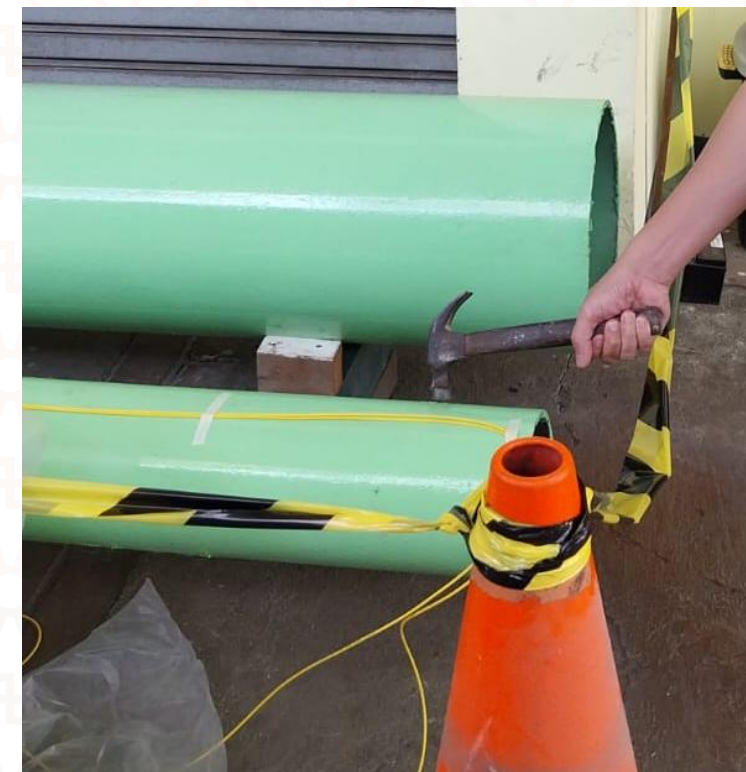


Hammer Position from Deep Learning AI



Hammer
Testing

Hammer Testing Performance: The system successfully identified and pinpointed impulsive hammer blows over the cable section (e.g., clearly segmenting impacts within the 100–330 m range).



Field-Scale Trial (Anyer, West Java)

50 km across Sunda Straits

